



An innovative bimodal computed tomography data-driven deep learning model for predicting aortic dissection: a multi-center study

Ziqian Li^{1#}, Lintao Chen^{1#}, Shengxuming Zhang^{2#}, Xiuming Zhang³, Jing Zhang¹, Mingliang Ying⁴, Jianyong Zhu⁵, Ruiyang Li⁶, Mingli Song², Zunlei Feng², Jianjun Zhang⁶, Wenjie Liang¹

¹Department of Radiology, the First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China; ²Department of Computer Science and Technology, Zhejiang University, Hangzhou, China; ³Department of Pathology, the First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China; ⁴Department of Pathology, Jinhua Central Hospital, Affiliated to Zhejiang University, Jinhua, China; ⁵Department of Emergency, the First Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China; ⁶Department of Radiology, Zhejiang Hospital, Affiliated to Zhejiang University, Hangzhou, China

Contributions: (I) Conception and design: Z Feng, Jianjun Zhang, W Liang; (II) Administrative support: W Liang; (III) Provision of study materials or patients: Z Li, X Zhang, L Chen, Jing Zhang, M Ying, J Zhu, R Li, Jianjun Zhang, W Liang; (IV) Collection and assembly of data: Z Li, X Zhang, L Chen, Jing Zhang, M Ying, J Zhu, R Li, Jianjun Zhang, W Liang; (V) Data analysis and interpretation: S Zhang, M Song, Z Feng; (VI) Manuscript writing: All authors; (VII) Final approval of manuscript: All authors.

[#]These authors contributed equally to this work.

Correspondence to: Wenjie Liang, MD. Department of Radiology, the First Affiliated Hospital, Zhejiang University School of Medicine, No. 79 Qingchun Road, Hangzhou 310003, China. Email: baduen@zju.edu.cn; Jianjun Zhang, MD. Department of Radiology, Zhejiang Hospital, Affiliated to Zhejiang University, No. 12 Lingyin Road, Hangzhou 310007, China. Email: zhangyp31113@163.com; Zunlei Feng, PhD. Department of Computer Science and Technology, Zhejiang University, No. 38 Zheda Road, Hangzhou 310027, China. Email: zunleifeng@zju.edu.cn.

Background: Aortic dissection (AD) is a lethal emergency requiring prompt diagnosis. Current computed tomography angiography (CTA)-based diagnosis requires contrast agents, which expends time, whereas existing deep learning (DL) models only support single-modality inputs [non-contrast computed tomography (CT) or CTA]. In this study, we propose a bimodal DL framework to independently process both types, enabling dual-path detection and improving diagnostic efficiency.

Methods: Patients who underwent non-contrast CT and CTA from February 2016 to September 2021 were retrospectively included from three institutions, including the First Affiliated Hospital, Zhejiang University School of Medicine (Center I), Zhejiang Hospital (Center II), and Yiwu Central Hospital (Center III). A two-stage DL model for predicting AD was developed. The first stage used an aorta detection network (AoDN) to localize the aorta in non-contrast CT or CTA images. Image patches that contained detected aorta were cut from CT images and combined to form an image patch sequence, which was inputted to an aortic dissection diagnosis network (ADDiN) to diagnose AD in the second stage. The following performances were assessed: aorta detection and diagnosis using average precision at the intersection over union threshold 0.5 (AP@0.5) and area under the receiver operating characteristic curve (AUC).

Results: The first cohort, comprising 102 patients (53±15 years, 80 men) from two institutions, was used for the AoDN, whereas the second cohort, consisting of 861 cases (55±15 years, 623 men) from three institutions, was used for the ADDiN. For the AD task, the AoDN achieved AP@0.5 99.14% on the non-contrast CT test set and 99.34% on the CTA test set, respectively. For the AD diagnosis task, the ADDiN obtained an AUCs of 0.98 on the non-contrast CT test set and 0.99 on the CTA test set.

Conclusions: The proposed bimodal CT data-driven DL model accurately diagnoses AD, facilitating prompt hospital diagnosis and treatment of AD.

Keywords: Aorta; aortic dissection (AD); computed tomography (CT); deep learning (DL)

Submitted Dec 11, 2024. Accepted for publication May 23, 2025. Published online Aug 18, 2025.

doi: 10.21037/qims-2024-2807

View this article at: <https://dx.doi.org/10.21037/qims-2024-2807>

Introduction

Aortic dissection (AD) is a life-threatening condition, representing the most common manifestation of acute aortic syndromes (1). The incidence of AD has steadily increased over the last 20 years, with an overall mortality rate reaching 27.4% (2,3). Prompt and accurate detection and timely surgical intervention remain critical in addressing the ongoing clinical challenge of diagnosing and treating AD (4).

The current consensus emphasizes the essential role of medical imaging in diagnosing AD (5). Chest radiography reveals essential clues for diagnosing AD, including mediastinal and aortic widening (6). Computed tomography angiography (CTA) facilitates accurate AD diagnosis by recognizing specific characteristics such as intimal tears and variable morphologies of false and true lumina (6). In addition, a detailed evaluation of AD can help to classify AD, guide the selection of therapeutic schemes, and predict patient prognosis. However, performing CTA for patients with specific conditions such as renal insufficiency, hyperthyroidism, or contrast allergy carries significant risks. Thus, the development of non-contrast computed tomography (CT)-based diagnostic methods for AD addresses an urgent clinical need to improve diagnostic accuracy in specific medical scenarios.

In recent years, artificial intelligence (AI) algorithms and deep learning (DL) models have been utilized to analyze medical imaging data. The quantitative features are deeply extracted and used to build predictive models for diagnosis, disease severity, and therapeutic assessment (7-11). This approach has been recognized as valuable, with DL models based on contrast-enhanced CT data enabling rapid AD identification (12,13). Several studies have also explored diagnostic models based on non-contrast CT data (14-16). However, current AI systems for AD diagnosis exhibit two fundamental limitations: (I) existing models fail to incorporate both CTA and non-contrast CT data; and (II) the inherent opacity of DL algorithms obscures decision-making processes, undermining clinical confidence. Our current study simultaneously addresses these challenges by developing a bimodal analytical framework augmented with explainable heatmap visualizations, thereby bridging the gap

between technical performance and clinical applicability. Our AI system also helps to reduce the emergency interpretation workload for radiologists and supports faster AD diagnosis.

In this study, we developed a bimodal CT-based deep learning model (BCT-DLM) comprising two components: the aorta detection network (AoDN) and the aortic dissection diagnosis network (ADDiN). The system integrates two dedicated modules: a non-contrast CT DL model (NCCT-DLM) for initial AD assessment in emergency or contrast-contraindicated cases, and a CTA DL model (CTA-DLM) for definitive diagnosis. The code and model implementation details for this study are publicly accessible via a GitHub repository (<https://github.com/zsxm1998/AorticDissection>). This dual-modality approach addresses critical limitations in current AD diagnostic pathways by improving screening efficacy, demonstrating significant clinical potential. We present this article in accordance with the TRIPOD+AI reporting checklist (available at <https://qims.amegroups.com/article/view/10.21037/qims-2024-2807/rc>).

Methods

Study sample

This retrospective multicenter study enrolled patients from three participating institutions, including the First Affiliated Hospital, Zhejiang University School of Medicine (Center I); Zhejiang Hospital (Center II); and Yiwu Central Hospital (Center III), between February 2016 and September 2021. Detailed scanning parameters and institutional information are provided in [Appendix 1](#). Case participants were defined as patients with a definitive diagnosis of AD confirmed by both clinical evaluation and CTA findings. Control participants consisted of individuals without AD who underwent non-contrast CT or CTA examinations for aortic evaluation during the same period (February 2016 to September 2021), with complete imaging data available for analysis. This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. The study was approved by Clinical Research Ethical Committee of the First Affiliated

Hospital, Zhejiang University School of Medicine (reference number: 2022 No. 892). All participating hospitals were informed of and agreed with the study. The requirement for written informed consent was waived because of the retrospective nature of the study.

The inclusion criteria were as follows: for the AD cohort: (I) confirmed AD diagnosis through CTA and clinical assessment. (II) Availability of preoperative non-contrast CT and/or CTA imaging data. For the non-AD cohort: (I) absence of AD, aortic aneurysm, aortitis, or intramural hematoma. (II) Availability of non-contrast CT and/or CTA imaging data. The exclusion criteria (applied to both cohorts) were as follows: (I) suboptimal image quality (e.g., significant motion artifacts). (II) Incomplete aortic coverage (defined as missing any segment between the supra-aortic arch region and iliac bifurcation). (III) Presence of intravascular aortic devices (e.g., stents). (IV) History of aortic surgical procedures prior to imaging acquisition.

For the AoDN cohort, we retrospectively collected 102 AD patients with non-contrast CT or CTA images in two institutions (Center I, III) from October 2016 to July 2021. As some patients had undergone only non-contrast CT or CTA images, we manually selected 88 patients for non-contrast CT network training and 90 for CTA network training. A flowchart of this study is depicted in *Figure 1A*. In addition, for the ADDiN cohort, we enrolled 900 cases in three institutions (Center I, II, III) from February 2016 to September 2021. Finally, we selected 861 cases and divided them into the training, validation, and test sets, as shown in *Figure 1B*. The technical roadmap of this study is as shown in *Figure 2*.

Construction of the AoDN

For the aorta detection module, manual annotations were performed on an initial cohort of 88 patients. These cases were selected based on their early availability and sufficient thoracic aorta coverage, which ensured consistent labeling. Due to the intensive nature of manual annotation, this subset was used to train and validate the AoDN. Additional detection annotations were not required for the classification task.

The initial step to facilitate automatic AD diagnosis involves locating the aorta on non-contrast CT/CTA images. Consequently, we developed the AoDN using DL techniques. Initially, we converted the Digital Imaging and Communications in Medicine (DICOM) images into 8-bit depth PNG grayscale format to facilitate AoDN

training. Using bounding boxes, we utilized the RadiAnt DICOM viewer software to label the aortic positions from the top of the aortic arch to the bifurcation. This labeled data was then utilized for training the aortic detection model. Two radiologists (Yubizhuo Wang and Hongbin Zhang, with more than 5 years of experience in emergency radiology) performed the annotation, and a senior radiologist (W.L.) reviewed and confirmed the annotations. Only specific keyframes were selected for labeling, and to complete the labeling of other unlabeled frames, we applied linear interpolation to determine the bounding box positions and sizes.

We selected YOLOv5 (17) as the aorta detection neural network. This architecture divides images into a grid system where each grid detects objects within its boundaries. Given that each image contained at most two aortas (ascending and descending), we optimized computation by reducing the number of anchors per grid from 3 to 1. We utilized pre-trained weights from the Common Objects in Context (COCO) dataset (18) to provide the model with essential image understanding capabilities.

Specifically, the AoDN employs the YOLOv5 architecture, trained with a batch size of 32, an initial learning rate of 0.01, and the stochastic gradient descent (SGD) optimizer (momentum = 0.937). Training proceeded for 100 epochs using a cosine annealing learning rate scheduler, with model initialization from COCO dataset-pretrained weights. These parameters have been added to [Table S1](#).

Construction of AD diagnostic network

We began by processing the non-contrast CT and CTA images of the ADDiN cohort. The slices corresponding to AD in the CTA sequence were labeled for each AD patient. These labeled slices were then aligned with the corresponding slices from the non-contrast CT scans to obtain labels for the non-contrast CT. Considering that AD typically manifests through specific imaging features across contiguous slices, including intimal tears and true/false lumina, the model for AD patients was trained and tested using only the labeled slices, whereas all aortic slices were utilized to train and test the model for negative participants.

The trained AoDN was utilized to identify the aortic regions in the dataset used for ADDiN training. The detected aortas were then segmented into image patches. Specific methods are provided in the Appendices. A three-dimensional (3D) convolutional neural network (CNN) was used to process the sequential aortic patches. A patch

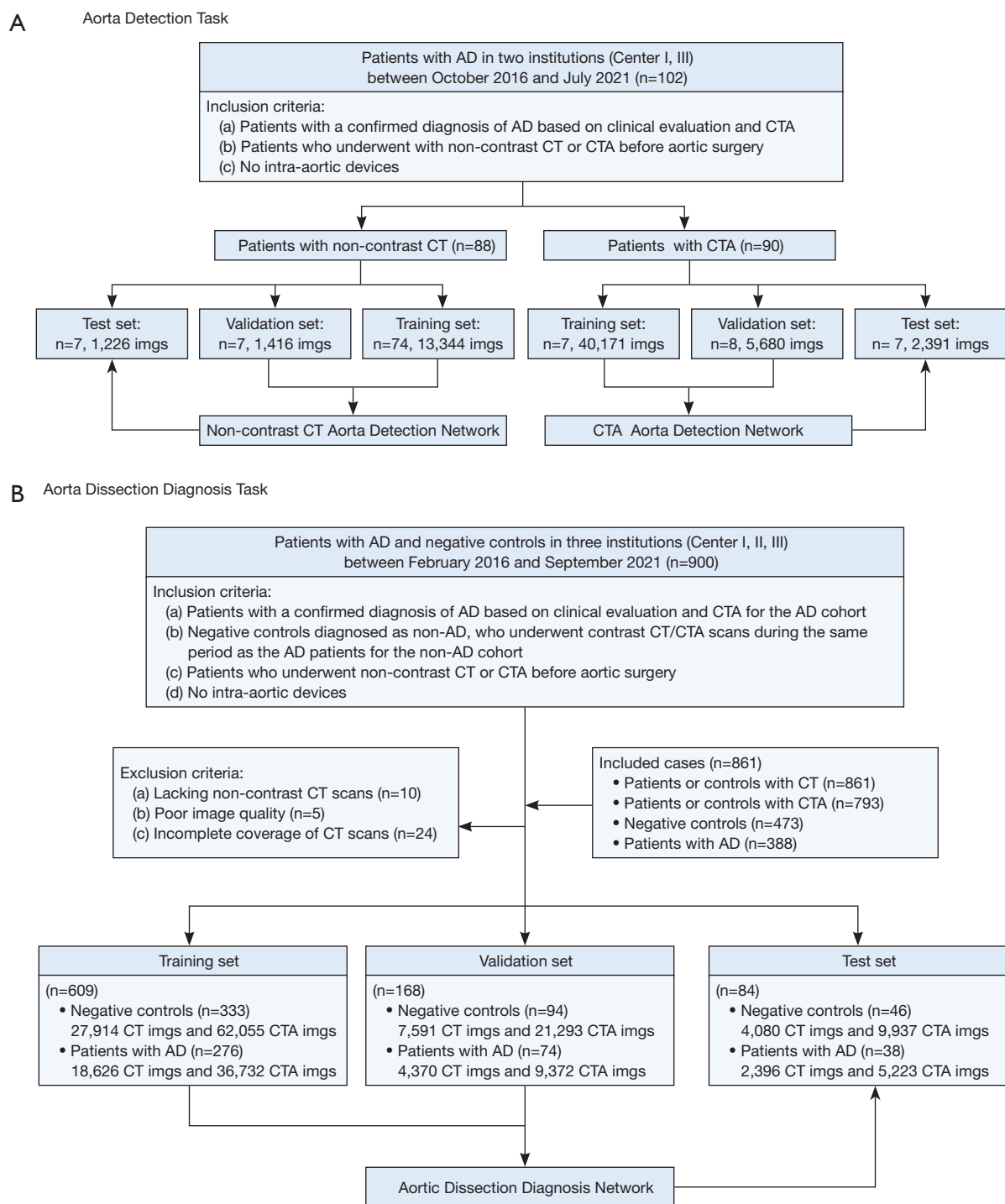


Figure 1 Flowchart of the study. (A) The aorta detection task included 102 patients from two institutions. Out of 102 patients, 88 were used to train the CT aorta detection network, and 90 were selected for training the CTA aorta detection network. (B) Data for the AD classification model were collected in three parts, consisting of training, validation, and test sets, with a total of 861 patients from three institutions. Center I, II, III: the First Affiliated Hospital Zhejiang University School of Medicine (Center I); Zhejiang Hospital (Center II); and Yiwu Central Hospital (Center III). AD, aortic dissection; CT, computed tomography; CTA, computed tomography angiography; imgs, images.

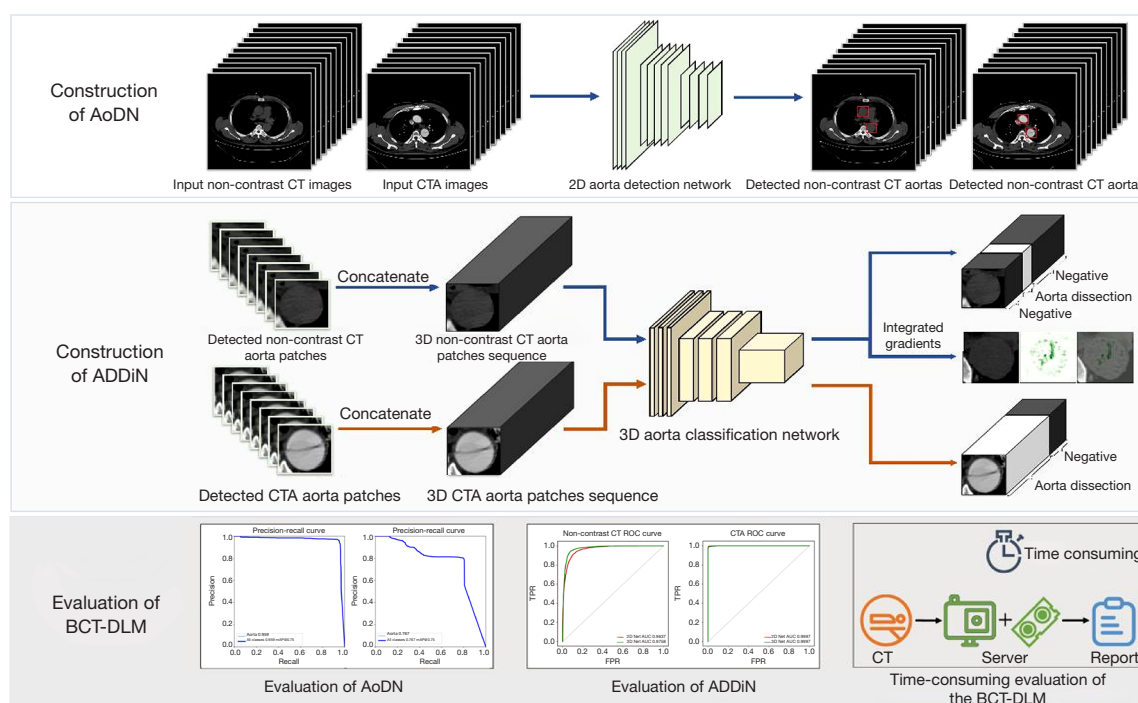


Figure 2 The proposed two-stage AI diagnosis framework for aortic dissection. The proposed BCT-DLM comprises an AoDN and an ADDiN. The AoDN, a 2D CNN, is used to locate the aorta in non-contrast CT or CTA images. Image patches containing the detected aorta are extracted from CT images and combined to form image patch sequences. These sequences are then input to the ADDiN, a 3D CNN, for diagnosing aortic dissection. We train two groups of AoDN and ADDiN for non-contrast CT and CTA separately, indicating that our framework can be applied to both non-contrast CT and CTA images independently. 2D, two-dimensional; 3D, three-dimensional; ADDiN, aortic dissection diagnosis network; AI, artificial intelligence; AoDN, aorta detection network; BCT-DLM, bimodal CT data-driven deep learning model; CNN, convolutional neural network; CT, computed tomography; CTA, computed tomography angiography.

sequence consisting of seven consecutive patches, including three patches preceding and three patches following the center patch, was employed as input for the 3D CNN to determine the category of the center patch. Each patch underwent the Sobel operator, and the resulting outputs were concatenated with the original patches to create an additional channel in the input image. In particular, the aortic patches were classified into two categories, negative and AD, using a 3D ResNet-34 architecture (19).

The model was trained using the Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 16, applying early stopping based on validation of the area under the receiver operating characteristic (ROC) curve (AUC). Training proceeded for a maximum of 60 epochs with a step-decay learning rate schedule ($\gamma = 0.1$ every 20 epochs). The model was initialized without pretrained weights. These parameters have been added to Table S1.

The presence of AD in an individual was determined by

considering a sufficient number of continuous aortic image patches classified as positive for AD by our classification model, taking advantage of the spatial continuity property of AD. The precise number of continuous aortic patches was determined through experimental analysis.

Visualization of BCT-DLM prediction

We used integrated gradients (IG) to visualize the critical regions in the input aortic image patches that the classification network relied on for predicting AD (20). For each input CT slice, we used a zero-valued baseline of the same shape as the input image. The attributions were computed by accumulating the gradients along a straight-line path between the baseline with the input through 100 integration steps. Each slice was processed independently to generate two-dimensional (2D) attribution maps, which were then reconstructed for 3D volumetric interpretation.

Table 1 Characteristics of the aorta detection dataset

Network	Dataset	Images	Patients	Men	Age (years)
AoDN	Non-contrast CT training set	13,344	74	61 (82.43)	54±14
	Non-contrast CT validation set	1,416	7	6 (85.71)	53±9
	Non-contrast CT test set	1,226	7	7 (100.00)	66±15
	CTA training set	40,171	75	59 (78.67)	51±14
	CTA validation set	5,680	8	6 (75.00)	54±10
	CTA test set	2,391	7	7 (100.00)	66±15
ADDiN	Non-contrast CT training set	46,540	609	436 (71.59)	55±16
	Non-contrast CT validation set	11,961	168	121 (72.02)	56±14
	Non-contrast CT test set	6,476	84	66 (78.57)	55±17
	CTA training set	98,787	556	402 (72.30)	55±16
	CTA validation set	30,665	158	117 (74.05)	56±14
	CTA test set	15,160	79	62 (78.48)	55±17

Data are presented as n, n (%) or mean ± standard deviation. ADDiN, aortic dissection diagnosis network; AoDN, aorta detection network; CT, computed tomography; CTA, computed tomography angiography.

To aid visualization, we normalized the attribution maps and overlaid them on the original CT images using heatmaps.

Statistical analysis

To define an individual as having AD, we used a certain threshold (T) of contiguous aortic patches predicted as AD by our classification model. The person-level classification performance of the model on non-contrast CT images was evaluated using different T values ranging from 1 to 20. Statistical significance was defined as $P < 0.05$. After training the AoDN, we assessed its performance on the test dataset using precision, recall, average precision at the intersection over union threshold 0.5 (AP@0.5), and AP@0.5:0.95 [average precision across intersection over union (IoU) thresholds 0.5 to 0.95]. The specific method for AP@0.5 calculation is provided in [Appendix 1](#). We evaluated AoDN's performance on the test datasets using the ROC curve and computed the AUC as the primary metric. Accuracy, sensitivity, and specificity were also determined for the test datasets.

Results

Characteristics of the study sample

A total of 102 patients (mean age: 53±15 years, 80 men, 22 women) were included in the AoDN cohort (*Figure 1A*).

In the ADDiN cohort, out of the 900 cases, 10 cases were excluded due to limited availability of non-contrast CT scans, five cases due to poor imaging quality, and 24 cases due to incomplete scan coverage. A total of 861 cases (mean age: 55±15 years, 623 men, 238 women) were included in the ADDiN cohort (*Figure 1B*). All relevant details can be found in *Table 1* and [Table S2](#).

Evaluation of the AoDN

We conducted five training iterations of our model to determine the mean and standard deviation of each discrimination index. The performance of the AoDN is presented in *Table 2*, and the precision-recall curves for non-contrast CT and CTA are shown in *Figure 3*. In non-contrast CT aorta detection, the model achieved excellent precision, recall, AP@0.5, and AP@0.5:0.95 scores of 97.98%, 97.88%, 99.14%, and 78.24%, respectively, on the test dataset. For CTA aorta detection, the precision, recall, AP@0.5, and AP@0.5:0.95 were 98.88%, 98.76%, 99.34%, and 73.88%, respectively, on the test dataset. The evaluation indicators of the CTA detection model did not demonstrate significant superiority over the non-contrast CT detection model.

Predictive performance of AD diagnostic network

We compared the performance of the 2D ResNet-34

Table 2 Performance metrics of the detection model

Set	Precision (%)	Recall (%)	AP@0.5 (%)	AP@0.5:0.95 (%)
Non-contrast CT test set	97.98±0.41	97.88±0.52	99.14±0.10	78.24±1.52
CTA test set	98.88±0.15	98.76±0.19	99.34±0.05	73.88±0.63

Data are presented as mean ± standard deviation. AP@0.5: average precision at the intersection over union threshold 0.5. AP@0.5:0.95: average precision across IoU thresholds 0.5 to 0.95. CT, computed tomography; CTA, computed tomography angiography.

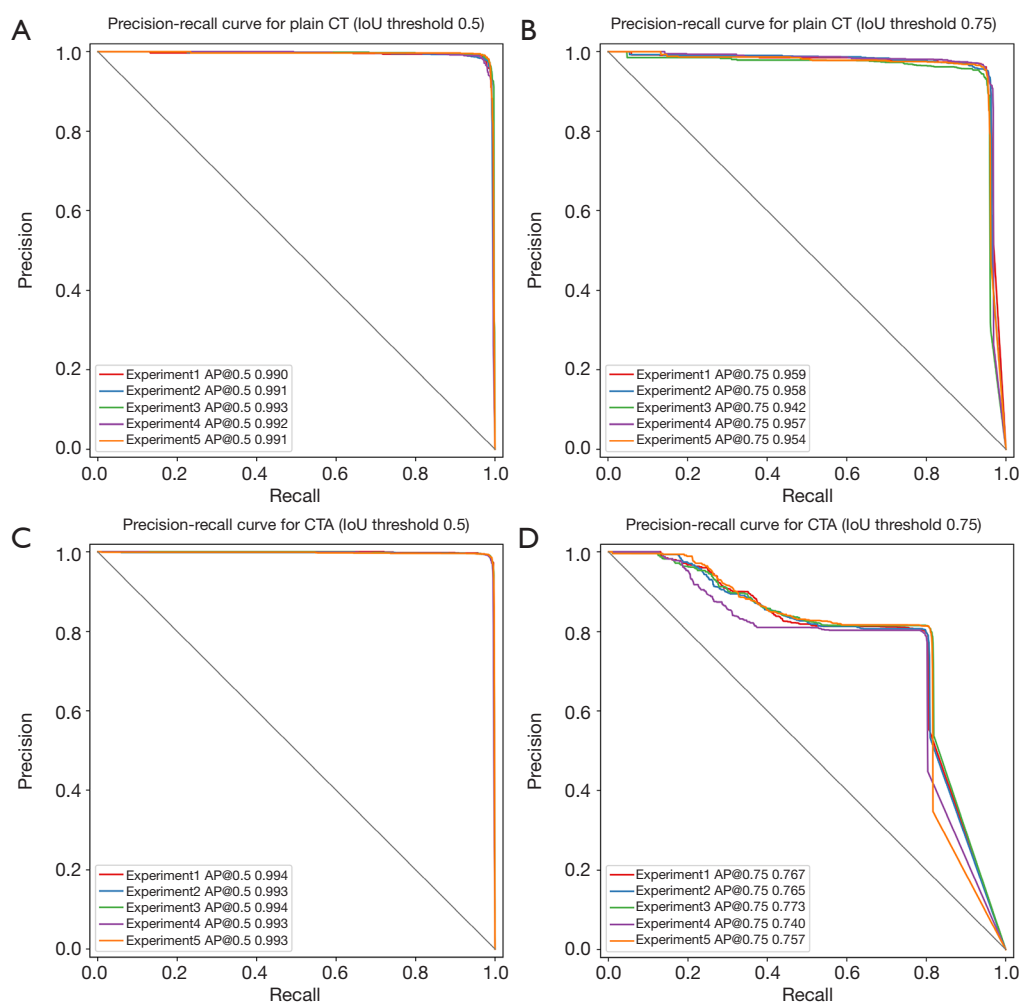


Figure 3 The precision-recall curve of the aorta detection model. (A) Precision-recall curve for plain CT (IoU threshold 0.5). (B) Precision-recall curve for plain CT (IoU threshold 0.75). (C) Precision-recall curve for CTA (IoU threshold 0.5). (D) Precision-recall curve for CTA (IoU threshold 0.75). AP@0.5: average precision at the intersection over union threshold 0.5. AP@0.75: average precision at IoU threshold 0.75. CT, computed tomography; CTA, computed tomography angiography; IoU, intersection over union.

and 3D ResNet-34 networks that were utilized on the test set. The ROC curves are depicted in *Figure 4*, and the metrics are presented in *Table 3*. The 2D ResNet-34 network achieved an AUC of 0.9637, accuracy of 89.83%, sensitivity of 90.76%, and specificity of 89.19% on non-

contrast CT. In contrast, the 3D ResNet-34 network achieved an AUC of 0.9767, accuracy of 92.75%, sensitivity of 94.47%, and specificity of 91.74%. All of these metrics of the 3D network significantly outperformed those of the 2D network (*t*-test, $P < 0.001$), indicating the superiority of

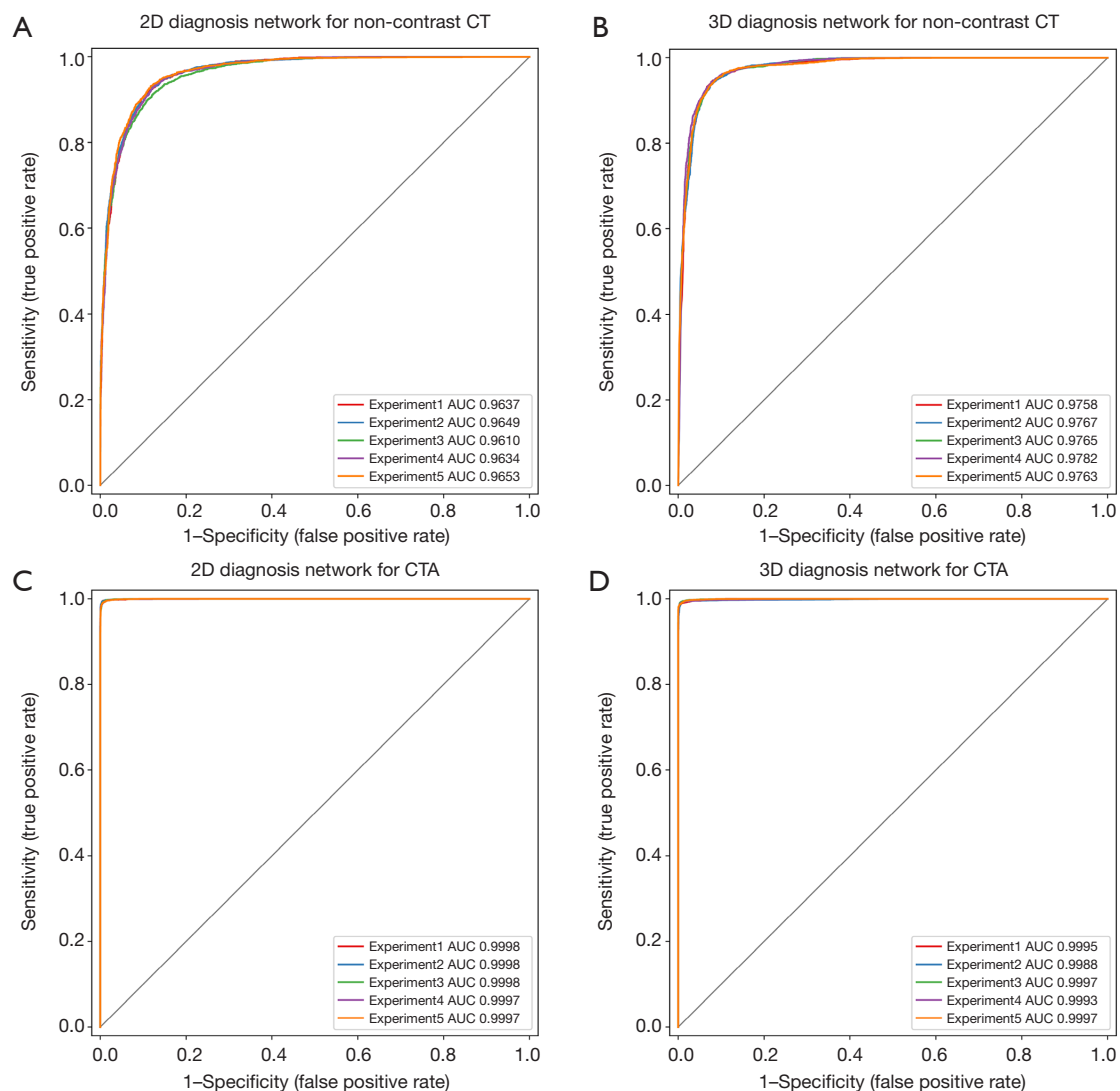


Figure 4 ROC curves of the aortic dissection classification model. (A) 2D diagnosis network for non-contrast CT. (B) 3D diagnosis network for non-contrast CT. (C) 2D diagnosis network for CTA. (D) 3D diagnosis network for CTA. 2D, two-dimensional; 3D, three-dimensional; AUC, area under the receiver operating characteristic curve; CT, computed tomography; CTA, computed tomography angiography; ROC, receiver operating characteristic.

the 3D classification network using continuous aortic image patches in AD classification on non-contrast CT images. However, both the 2D and 3D networks demonstrated excellent performance on the CTA test set with AUC, accuracy, sensitivity, and specificity close to 1, and no significant differences in partial metrics (AUC, accuracy) (t -test, $P > 0.05$), as shown in Figure 4 and Table 3.

The highest AUC of 0.9501 was achieved when T was 10, with corresponding accuracy, sensitivity, and specificity of 90.71%, 95.26%, and 86.96%, respectively. As T increased from 1 to 20, sensitivity increased while

specificity decreased. AUC and accuracy initially increased but eventually reached a plateau or decreased with further increase in T. The relevant results are presented in Table 4.

Time-consuming evaluation of the BCT-DLM

We assessed the computational time of our proposed framework on the test dataset. The evaluation involved various steps, including DICOM to PNG conversion, aortic detection, patch cutting and concatenation, and aortic sequence classification. The average processing time per

Table 3 Performance metrics of the classification model

Metric	Non-contrast CT test set			CTA test set		
	2D network	3D network	P value	2D network	3D network	P value
AUC	0.9637±0.0015	0.9767±0.0008	<0.001	0.9998±0.0000	0.9994±0.0003	0.07
Accuracy (%)	89.83±0.25	92.75±0.15	<0.001	99.34±0.08	99.24±0.11	0.16
Sensitivity (%)	90.76±1.34	94.47±0.27	<0.001	98.84±0.20	98.11±0.32	0.005
Specificity (%)	89.19±0.53	91.74±0.32	<0.001	99.65±0.09	99.83±0.05	0.01

Data are presented as mean ± standard deviation. 2D, two-dimensional; 3D, three-dimensional; AUC, area under the receiver operating characteristic curve; CT, computed tomography; CTA, computed tomography angiography.

Table 4 Performance metrics of individual-level classification on non-contrast CT test set

T	AUC	Accuracy (%)	Sensitivity (%)	Specificity (%)
1	0.7848±0.0340	61.19±2.78	100.00±0.00	29.13±5.07
2	0.8274±0.0315	68.10±3.05	100.00±0.00	41.74±5.57
3	0.8590±0.0270	73.10±3.81	99.47±1.05	51.30±7.09
4	0.8781±0.0093	78.33±1.90	98.42±1.29	61.74±2.95
5	0.8987±0.0186	81.43±1.21	97.89±1.05	67.83±2.13
6	0.9204±0.0116	85.71±1.30	97.89±1.05	75.65±2.54
7	0.9306±0.0164	88.33±1.39	97.37±0.00	80.87±2.54
8	0.9466±0.0074	90.00±1.21	96.84±1.05	84.35±1.63
9	0.9480±0.0094	89.76±1.61	95.79±1.29	84.78±1.94
10	0.9501±0.0065	90.71±1.75	95.26±1.05	86.96±2.38
11	0.9465±0.0087	90.48±1.68	93.68±1.29	87.83±2.22
12	0.9445±0.0125	89.52±2.76	91.05±3.94	88.26±2.61
13	0.9459±0.0092	90.00±2.45	90.53±3.57	89.57±2.13
14	0.9459±0.0083	90.71±1.90	90.00±3.87	91.30±2.38
15	0.9435±0.0077	90.71±1.58	88.42±2.68	92.61±1.06
16	0.9379±0.0121	89.76±1.21	86.32±1.97	92.61±1.06
17	0.9404±0.0087	89.05±0.89	84.74±1.05	92.61±1.06
18	0.9409±0.0066	89.05±0.89	84.21±0.00	93.04±1.63
19	0.9323±0.0182	89.29±0.75	84.21±0.00	93.48±1.37
20	0.9271±0.0120	89.29±0.75	84.21±0.00	93.48±1.37

Data are presented as mean ± standard deviation. AUC, area under the receiver operating characteristic curve; CT, computed tomography; T, threshold.

CT image was 0.1028 seconds, enabling the framework to handle 9.73 CT images per second. A CT sequence with 300 slices could be processed automatically in approximately 30.83 seconds, significantly reducing the workload of radiologists and accelerating the diagnostic process. The workstation configuration we used in the experiment is

provided in [Appendix 1](#).

Visualization of BCT-DLM prediction results

DL techniques often lack interpretability, which is crucial in medical image analysis for reliable and interpretable

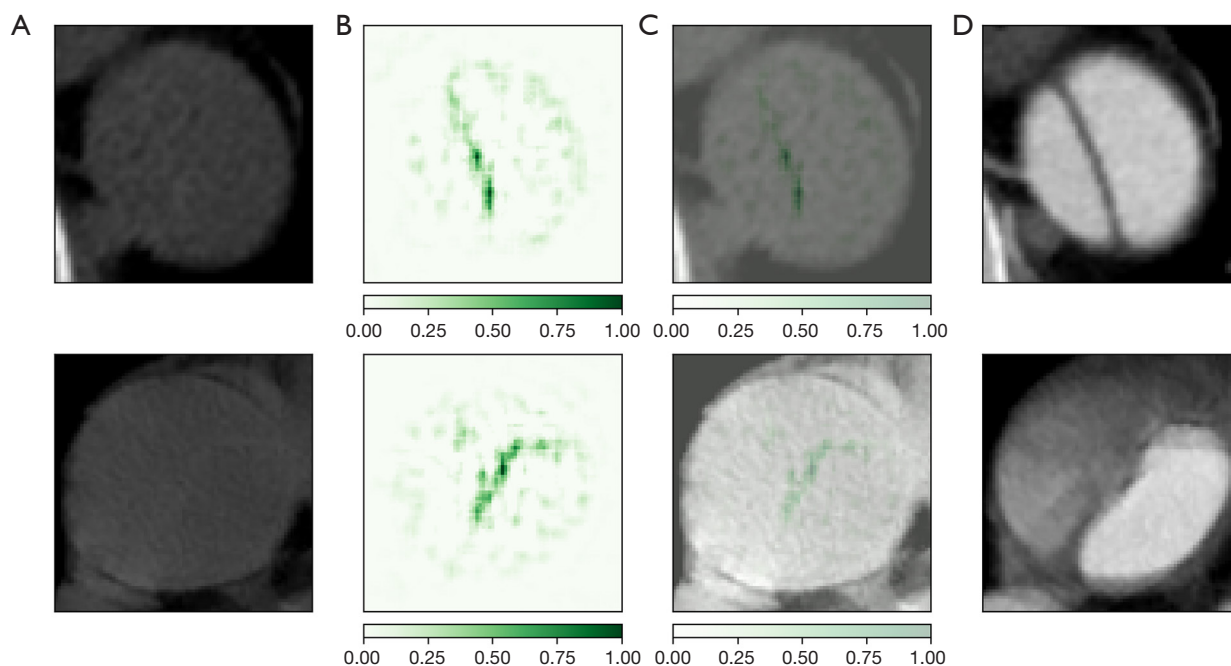


Figure 5 Visualization results of aortic dissection classification. (A) Non-contrast CT patch of the aorta. (B) Heatmap showing the importance score of the CT patch. (C) Fused image of the heatmap and the CT patch. (D) Corresponding CTA patch of the aorta. CT, computed tomography; CTA, computed tomography angiography.

predictions. To address this challenge, we employed IG to visualize the critical areas of input aortic image patches that the neural network relies on for AD prediction.

Figure 5 showcases specific visualized results, consisting of four images per sample. The first image displays the CT patch of the aorta, the second presents a heatmap indicating the importance scores of the CT patch, the third shows a blended heatmap with the CT patch, and the last image exhibits the corresponding CTA patch of the aorta. The importance scores range from 0 to 1, with higher scores indicating greater significance of the corresponding pixels in the model's AD diagnosis. The diagnostic model primarily focuses on intimal tears for AD classification.

A detailed description of the implementation is provided in Appendix 1, and additional visualization examples are presented in Figure S1 to further demonstrate the reliability of the explanation results.

Discussion

We developed a novel two-stage DL model utilizing bimodal CT data (CTA or CT images) to predict AD. The first network locates the aorta in CT images, whereas the

second network classifies the detected aorta as negative or AD. Our AI model for AD diagnosis offers three key advantages: firstly, it demonstrates the diagnostic capability of the AI model using non-contrast CT data. Secondly, it enables quick and accurate diagnosis, expediting the in-hospital diagnostic process and potentially reducing in-hospital mortality from AD. Thirdly, the results of our AI model are presented visually for enhanced comprehension by clinicians.

Our study demonstrates unique advantages across multiple dimensions. Unlike single-center studies (15,21), we integrated bulk CT data from three independent institutions with varying scanner models and acquisition parameters, significantly enhancing the model's robustness and generalizability across clinical environments. Previous studies had developed AD prediction models exclusively for either non-contrast CT (14,15,22,23) or contrast-enhanced CT (12,13,21,24) data; our bimodal framework enables independent analysis using either imaging modality, substantially improving clinical utility. By employing a 3D ResNet-34 network to analyze aortic patch sequences and leverage spatial continuity, we achieved superior accuracy, whereas their 2D Xception-based approach processing

individual slices may miss longitudinal pathological features (15). Notably, although previous studies reported stronger predictive efficacy in contrast-enhanced CT (23,24), our model using non-contrast CT demonstrated similarly high accuracy (92.75%), sensitivity (94.47%), and specificity (91.74%). More detailed comparisons of parameters between previous studies and the present study are provided in [Table S3](#).

To define an individual as having AD, we used a specific T of contiguous aortic patches predicted as AD by our classification model. The selection of T =10 for person-level AD diagnosis reflects a balance between sensitivity and specificity in our cohort, leveraging spatial continuity to mitigate single-slice false positives. Although this value achieved the highest AUC (0.9501) on our test set, its generalizability may be influenced by dataset-specific factors such as slice thickness or AD distribution patterns. Clinically, T is not fixed but serves as an adjustable parameter. Lowering T prioritizes sensitivity in high-risk settings, whereas increasing T reduces false alarms in stable populations. Future multicenter studies are warranted to validate this threshold across diverse imaging protocols and patient demographics. Importantly, our framework's flexibility ensures adaptability to real-world clinical needs, where rapid triage and diagnostic accuracy must coexist.

The aorta detection model achieved strong AP@0.5 performance for both non-contrast CT and CTA, but showed decreased AP@0.75 precision for CTA. This difference primarily results from CTA's unique challenges: contrast-enhanced vessels display complex overlapping structures and dynamic artifacts that hinder precise localization, whereas abrupt intensity changes create boundary ambiguity where even correct detections may slightly deviate from annotations. Such factors disproportionately affect strict IoU =0.75 evaluation since small spatial mismatches substantially lower scores. Additional variability from inter-observer labeling differences and typically conservative CTA annotations further exacerbates the precision challenge. Although the model performs comparably across modalities at moderate IoU thresholds, its reduced CTA precision at higher thresholds reflects the modality's greater anatomical complexity compared to non-contrast CT. Future enhancements should focus on adaptive localization algorithms to improve high-precision detection in complex contrast-enhanced imaging.

DL techniques often lack interpretability, a critical requirement in medical image analysis for reliable clinical

predictions. To address this limitation, we implemented IG to identify diagnostically significant regions within aortic image patches that contribute to AD predictions (20). Unlike gradient-based methods such as gradient-weighted class activation mapping (Grad-CAM) that depend on intermediate feature maps and demonstrate limited architectural generalizability, IG operates directly on input data without structural modifications. This approach enhances robustness, particularly for deep networks containing non-standard layers. Moreover, IG's suitability for medical 3D volumetric analysis stems from its generation of spatially consistent attribution maps, enabling reliable tracking of clinically relevant features across sequential CT slices. Comparative analysis of CTA versus non-contrast CT heatmaps demonstrated the model's focus on intimal tears—the key diagnostic feature used by radiologists in CTA-based AD assessment. This interpretable framework effectively bridges the gap between AI's inherent opacity and clinically actionable insights, overcoming the limitations of conventional diagnostic approaches.

Our study has several limitations that need to be acknowledged. First, the limited sample size may have restricted thorough analysis of subgroups (e.g., atypical AD cases), and although cross-institutional validation was partially addressed, strict segregation by institution was not implemented, potentially affecting external validity. Second, although our model can localize the aorta for patch extraction and classification, it does not provide specific anatomical region localization (e.g., aortic arch) or segment-specific anatomical characterization for clinical AD typing. Third, reliance on annotated slices for training may reduce sensitivity to atypical presentations in unannotated slices, which could be addressed through weakly supervised learning approaches. Fourth, the model processes CT data in isolation without incorporating clinically relevant variables (e.g., biomarkers, medical history), and requires real-world validation against radiologist performance. Lastly, our model currently lacks real-world comparative studies. In future prospective research, we will compare the diagnostic performance between radiologists and the AI model based on non-contrast CT data to further validate its clinical utility.

Conclusions

Our study utilized CT data and DL algorithms to develop an effective bimodal data-driven AoDN and ADDiN for AD prediction. The visualization results demonstrate that

the ADDiN accurately predicts AD in non-enhanced CT by specifically targeting the intimal tear region. To enhance the robustness of our model for clinical use, further optimization is required by acquiring more CT data and implementing a prospective design.

Acknowledgments

None.

Footnote

Reporting Checklist: The authors have completed the TRIPOD+AI reporting checklist. Available at <https://qims.amegroups.com/article/view/10.21037/qims-2024-2807/rc>

Data Sharing Statement: Available at <https://qims.amegroups.com/article/view/10.21037/qims-2024-2807/dss>

Funding: This work was supported by the 2024 Zhejiang Provincial Medical and Health Science and Technology Program (grant No. 2024KY1756).

Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at <https://qims.amegroups.com/article/view/10.21037/qims-2024-2807/coif>). The authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments. The study was approved by Clinical Research Ethical Committee of the First Affiliated Hospital, Zhejiang University School of Medicine (reference number: 2022 No. 892). All participating hospitals were informed of and agreed with the study. The requirement for written informed consent was waived because of the retrospective nature of the study.

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Cite this article as: Li Z, Chen L, Zhang S, Zhang X, Zhang J, Ying M, Zhu J, Li R, Song M, Feng Z, Zhang J, Liang W. An innovative bimodal computed tomography data-driven deep learning model for predicting aortic dissection: a multi-center study. *Quant Imaging Med Surg* 2025;15(9):8359-8371. doi: 10.21037/qims-2024-2807

Appendix 1

Scanning Parameters and Names of Institutions

Cases from Center I (the First Affiliated Hospital, Zhejiang University School of Medicine) underwent non-contrast CT or CTA with Philips Brilliance iCT 256 (Philips Health Care, Cleveland, USA). The scanning parameters were as follows: tube voltage 120kv, tube current 300mA, matrix size 512×512 , pitch 0.915, scanning speed 0.4s, scanning layer thickness 5mm, reconstruction layer thickness: 1mm. Scanning range is from 1cm above the aortic arch to the bifurcation of iliac artery. Iohexol contrast agent was used, and 1.5-2ml/kg of contrast agent was injected through the forearm through a high-pressure syringe at a flow rate of 4.5-5ml/s.

Cases from Center II (Zhejiang Hospital, affiliated to Zhejiang University School of Medicine) underwent non-contrast CT or CTA with Siemens Somatom Force. The scanning parameters were as follows: tube voltage 100kv, tube current automatic regulation (reference value 288mA), matrix size 512×512 , pitch 1.5, scanning speed 0.5s, scanning layer thickness 5mm, reconstruction layer thickness: 0.6mm. Scanning range is from 1cm above the aortic arch to the bifurcation of iliac artery. Iohexol contrast agent was used, and 1.5-2ml/kg of contrast agent was injected through the forearm through a high-pressure syringe at a flow rate of 4.5ml/s.

Cases from Center III (Yiwu central hospital, affiliated to Wenzhou Medical University) underwent non-contrast CT or CTA with GE Optima CT 680 (General Electric Healthcare, Chicago, IL, USA). The scanning parameters were as follows: tube voltage 120kv, tube current 300mA, matrix size 512×512 , pitch 1.375, scanning speed 0.6s, scanning layer thickness 5mm, reconstruction layer thickness: 0.625mm. Scanning range is from 1cm above the aortic arch to the bifurcation of iliac artery. Iohexol contrast agent was used, and 1.5-2ml/kg of contrast agent was injected through the forearm through a high-pressure syringe at a flow rate of 4.5ml/s.

Creating Sequences of Aorta Patches for Construction of AD Diagnostic Network

First, intra-slice aorta localization assigns bounding boxes in axial CT slices containing both ascending and descending aorta (e.g., at the aortic arch level) based on anatomical relationships: the ascending aorta positioned proximal part (cardiac proximity) and descending aorta located distal part

(spinal proximity).

Second, inter-slice continuity verification maintains anatomical consistency through Intersection over Union (IoU) calculations between adjacent slices. For each bounding box in slice n , IoU values are computed against all candidate boxes in slice $n+1$, with pairs demonstrating maximum IoU (>0.7 threshold) linked as continuous aortic segments.

Finally, aortic patch processing involves cropping detected regions, resizing to 64×64 pixels, and axial stacking to construct 3D sequences (e.g., 7 consecutive slices) for subsequent 3D ResNet-34 network analysis.

Statistical Analysis

The AP@.5 was calculated as follows. First, its confidence scores sorted the predicted boxes in descending order. Then the precision and recall were calculated for each predicted box, as the threshold for matching the predicted and ground truth boxes used was set to 0.5 IoU. Then, the precision-recall curve was calculated by plotting the precision-recall pairs at various confidence thresholds. The area under the precision-recall curve was then calculated as the value of AP@.5. The AP@.5:.95 is the average of AP at IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05.

Implementation Details of Integrated Gradients

The Integrated Gradients (IG) method was implemented according to the definition by Sundararajan *et al.* (19). For each input CT slice, we used a zero-valued baseline with the same dimensions as the input image. The attributions were computed by accumulating the gradients along a straight-line path between the baseline and the input through 100 integration steps. Each slice was processed independently to generate 2D attribution maps, which were then reconstructed for 3D volumetric interpretation.

To enhance visualization, we normalized the attribution maps and overlaid them on the original CT images as heatmaps. *Figure S1* presents representative examples of these attributions. These visualizations reveal the model's focus areas and validate its attention to clinically relevant regions.

Workstation Specifications

The model was trained on a Dell 7920 workstation equipped with dual Intel Xeon 4215R processors, 128 GB

RAM, and an NVIDIA Quadro P6000 GPU.

Comparison of Atherosclerosis Between AD and Non-AD Groups

The prevalence of atherosclerosis is summarized in *Table S2*. In the ADDiN cohort, atherosclerosis were present in 60.9%

(288/473) of AD patients versus 55.9% (217/388) of non-AD controls. Similarly, in the AoDN cohort, atherosclerosis was in 58.0% (51/88) of non-contrast CT cases and 57.8% (52/90) of CTA cases. Chi-square analysis showed no significant differences in atherosclerosis prevalence between AD and non-AD groups (ADDiN: $p=0.12$) or between imaging modalities (AoDN: $p>0.20$).

Table S1 Training Hyper-parameters for AoDN and ADDiN

Network	AoDN	ADDiN
Architecture	YOLOv5	ResNet-34
Pretrained Weights	No Pretrain	No Pretrain
Optimizer	SGD (momentum = 0.937)	Adam
Learning Rate	0.01	1e-4
Batch Size	32	16
Epochs	100	Max 60 (early stopping applied)
Learning Rate Schedule	Cosine Annealing	Step Decay ($\gamma=0.1$ every 20 epochs)

Abbreviations: AoDN, aorta detection network; ADDiN, aortic dissection diagnosis network.

Table S2 Comparison of Atherosclerosis Between AD and Non-AD Groups

Network	Cohort	Cases	Cases with atherosclerosis
AoDN	Non-contrast CT set	88	51
	CTA set	90	52
ADDiN	the AD cohort	473	288
	the non-AD cohort	388	217

Abbreviations: AoDN, aorta detection network; ADDiN, aortic dissection diagnosis network.

Table S3

Summary of Previously Published Studies Pertaining to Automated Detection Solutions for AD

Parameter	Harris <i>et al.</i> , 2019 (12)	Yun <i>et al.</i> , 2020 (13)	Hata <i>et al.</i> , 2021 (15)	Xiong <i>et al.</i> , 2022 (14)	Huang <i>et al.</i> , 2022 (21)	Yi <i>et al.</i> , 2022 (22)	Cheng <i>et al.</i> , 2024 (23)	Laletin <i>et al.</i> , 2024 (24)	Current study
Types of studies	Retrospective	Retrospective	Retrospective	Retrospective	Retrospective	Retrospective	Retrospective	Retrospective	Retrospective
Institutions	Multicenter	One center	One center	Multicenter	One center	Multicenter	Multicenter	Multicenter	Multicenter
Medical imaging modalities	CTA	CTA	NCCT	NCCT	CTA	NCCT	NCCT	CTA	CTA & NCCT
Sample size	112 AD patients (34,577 cases)	43 AD patients (88 cases)	85 AD patients (170 cases)	93 AD patients (237 cases)	51 patients (298 cases)	185 AD patients (452 cases)	160 AD patients (320 cases)	137 AD patients (1303 cases)	388 AD patients (861 cases)
Architecture of CNN	5-layer CNN	DenseNet121	CNN Xception	3D nnU-Net	attention U-net and ResNeXt	2.5D U-net, ResNet34	The 3D full-resolution U-Net	2.5D U-Net	the YOLOv5, 3D ResNet-34
Model	2D	2D	2D	3D	3D	3D	3D	3D	3D
Performance comparison	Sensitivity	87.8%	82.2%	91.8%	93.8%	94.7%	86.2%	91.6%	94.5%
	Specificity	96.0%	96.8%	88.2%	72.5%	95.4%	92.3%	95.6%	91.7%
	Accuracy	96.9%	96.8%	90.0%	83.1%	96.6%	89.7%	93.8%	92.8%
Key Contributions	Aortic rupture analysis	Morphology-based ROI extraction	Grad-CAM-based algorithm interpretation	NCCT to CTA conversion	Type A/B classification	Expert comparison (3 readers)	Segment the true and false lumens in aortic NCCT	Algorithmic screening, classification & prioritization	IG-based visualization of key aortic regions

Abbreviations:

CTA, computed tomography angiography; AoDN, aorta detection network; ADDiN, aortic dissection diagnosis network; IG, integrated gradients; Grad-CAM, gradient-weighted class activation mapping; NCCT, non-contrast CT.

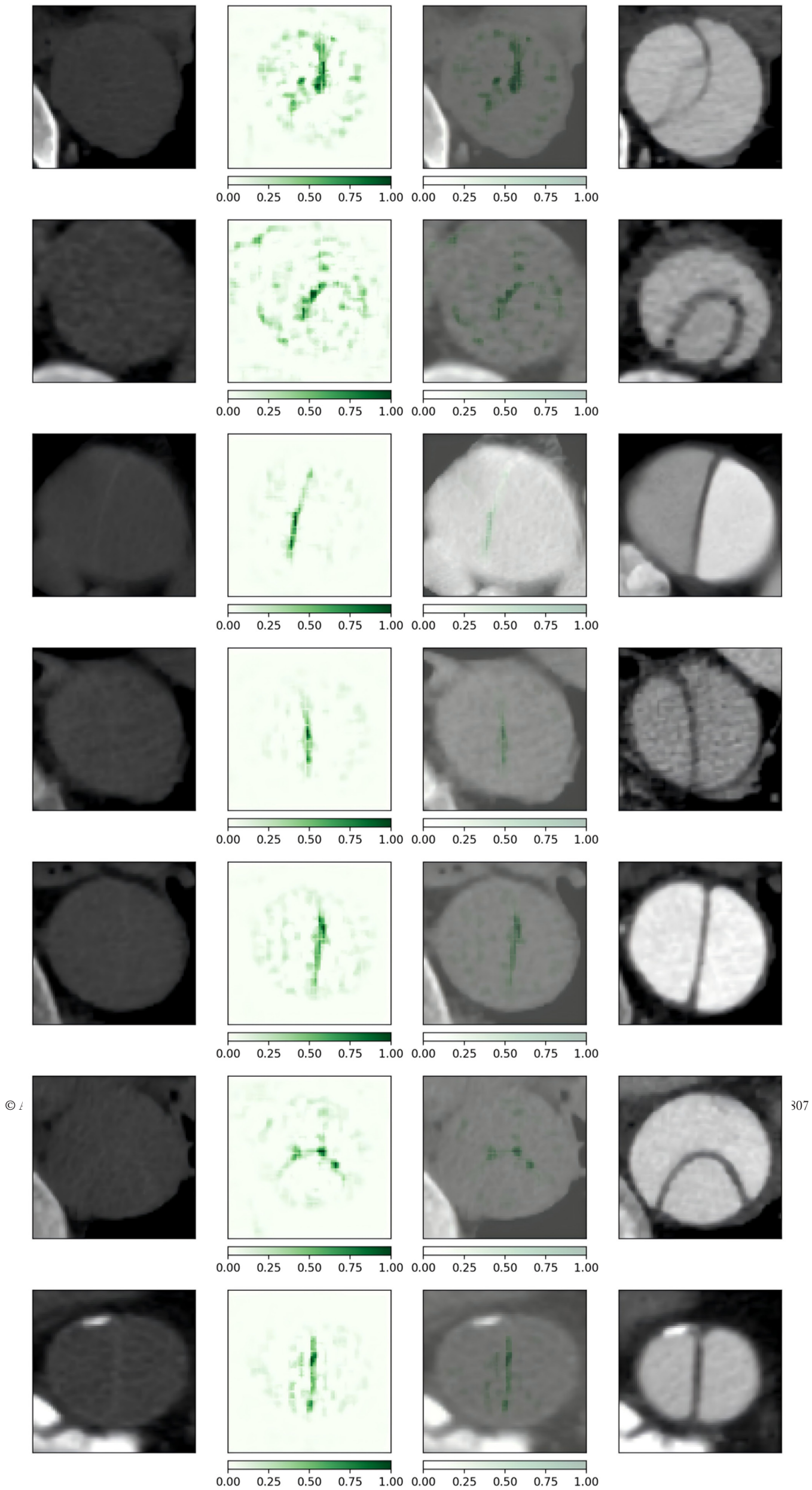


Figure S1 Visualization results of aortic dissection classification.